Give a Hand to Audits

Today, hand-held devices are smaller and easier to use and can be extended very easily with a wide variety of both software and hardware add-ons.

First and foremost, any device for audits needs to be portable and easy to use.

by Bill Shadish

The capability now exists for auditors to inspect a home; enter data; determine (by running local calculations) what measures are needed to increase the energy efficiency of the home; create a printed paper invoice/order for the customer; and digitally capture the customer's written signature of acceptance. An order for the materials can be placed into the back-end system and the date for doing the work determined with the customer—and this entire process can be completed in a single visit. But this is old news to some of you readers.

I want to share with you some of the more advanced things that can be added to a personal digital assistant (PDA) to make your time in the field even more productive. However, let's start by taking you through the minimum that you should demand when developing an audit application to run on hand-held devices.

Utilities, contracting firms, energy auditors, and general home inspectors throughout the United States perform audits using hand-held devices (PDAs). The technology of automating energy audits has grown a great deal since I first started to work with PDAs in 1997, and especially during the several years that I have been principal of Fundamental Objects. In the old days, a heavy, battery-draining block of plastic with a grayscale display and with no easy Internet connectivity, was the only available tool for automating audits. But even these tools made life a lot easier for those who had done everything with paper and pencil before that.

Today, not only are hand-held devices much smaller and easier to use, but they can be extended very easily with a wide variety of both software and hardware add-ons. This includes everything from blower door test data hooks and the ability to grab device temperatures automatically, to wireless earphones and wireless keyboards. And of course, Internet connectivity is widely available now, as well.

Table 1. Hardware Choices

Platform	Pros	Cons
PDA/ hand-helds	Small	Small display
	Low weight (6 oz)	
	Relatively low cost ($250–$600)	
	Wide variety of software tools	
	Runs select Windows tools, like Excel, Word	
	Can be carried in a pocket	
	Can include a phone	
	Can include a voice recorder	
	Can include a camera	
Traditional laptop/ notebook	Uses Windows tools	Weight (4-7 lb)
	Large display	No signatures, drawing
		Need 2 devices for phone
		Cost ($700-$2,600)
Tablet PC	Touch screen in a laptop-like format	Weight (3–6 lb)
		Need 2 devices for phone
		Generally underpowered compared to laptops
		Cost ($2,400–$5,000)
Smartphone, phone	Small	No touch screen
	Low weight	No standards for OS
	Can be carried in a pocket	Incompatible toolsets for development
	Can be used as a phone	Very small display

The Basic Choices

When considering what platform to use when automating your audits, there are several ways to go (see Table 1). The choices are based largely on how mobile you want the device to be. Almost all of the customers that we have worked with over the years at Fundamental Objects have gone the PDA route. Usually this includes some supporting Web site pieces as part of the overall solution. The idea is to provide the auditor or inspector with the smallest, lightest, easiest-to-

Because PDAs are small, you will want to get the largest, brightest screen.

PDAs Versus Smart Phones

One thing to be aware of is that Smartphones (a loose marketing term that is misapplied on some devices) are telephones first and foremost. Their display is smaller than a full PDA display, and the Smartphone display is not touch screen. That is, it cannot be operated with a stylus or fingertip, but only via the phone's keypad. The new iPhones are an exception; they have a relatively large touch screen display. The keypad is generally only a standard phone numeric keypad and is a lot more awkward to enter data with than a QWERTY-style keyboard.

Note: A Smartphone generally uses Java for programming, and the program is often tightly bound to the device. A Java program written for one phone may not work on others. PDAs are generally either PalmOS or Windows Mobile. Many of the audit tools that were written for earlier versions of these operating systems work today. Now before the Java folks come down on me, I will note that we have tried to take software programs from one phone platform and register them on others. The time and cost to do so is not small, nor is the labor involved.

hold device possible. Recently, the convergence of a phone and a PDA into a single device adds the benefit of reducing the things the auditor has to carry around (and possibly to drop, break, or lose) by one.

There are a lot of PDAs on the market. Mix in Smartphones and phones that run programs and there is certainly plenty to choose from (see "PDAs Versus Smart Phones"). First and foremost, the tool has to be easy to use. That is, it has to be easy to operate both the keys and any associated hardware gizmos (such as the battery, charger, stylus, and flip-out keyboard) on the PDA. It also has to be easy to run the audit software that you use. Note that this means easy to use when conducting audits. A cell phone might be very easy to operate—but it cannot run your audit software or print a report in the field. Similarly, a laptop might have all of the features that you need, but carrying eight floppy pounds around while inspecting a furnace may not be your idea of fun.

Secondly, the software has to be compatible with what you use today. If you manage a company that has Windows running on all of the desktops, then being able to run Excel and Word on the PDA makes a lot of sense.

Third, you have to be able to get data into and back out of the PDA easily. Data includes the audits. Data also includes the resulting reports associated with the audits. Other things that you will want to get into and out of your PDA easily are e-mail, documents, and perhaps data from external devices, such as temperature-measuring gauges or radio frequency identification (RFID) tags (which are discussed below).

Here are some more important features to look for:

High-Resolution Screen

PDAs are small. That is their strength. That is also their weakness. Because the device is small, you will want to get the largest, brightest screen, with the highest resolution that you can. The last point is important, because while two screens may be the same size, they may operate at different resolutions, and the one with the lower resolution might show significantly less information than the other.

Tappy Software

"Tappy" is not really a word, but I use it to define how well the actual audit software is set up to work for you. You will want most, if not all of the data entry items to be based on drop-down lists, checkboxes, radio buttons, or pick lists. The less auditors are actually typing (or thumbing in, as they would be on PDAs), the better the interface is. This reduces keying and rekeying errors, and reduces the time it takes to conduct an audit.

There are rules of thumb that determine when to use each type of input. For example, if something is just an either/or choice, like deciding if the water heater is already wrapped or not, then a checkbox works best. If there are more than three choices, then you probably want a drop-down list or a pick list if you have the space available on a screen. Either of these options keeps the auditor from having to key in this information; which is pretty much sure to result in an entry at least a little different from entries made by other auditors keying in the same exact thing.

Ease of Navigation

It should be easy for the user to navigate back and forth through the audit. Does the user have to complete all of the information on a page before he or she can save it? Can the entire audit be saved before it is finished, so the user can return to it later? Can the user operate the PDA audit with a fingertip, or is a stylus required? If a keyboard is required, then the design of the PDA is probably not well thought-out. You want to be as close to one-handed operation as possible. Remember—the other hand is going to be holding equipment, insulation, and so on.

How do you want to connect your PDAs? You can have your auditors work detached or wirelessly. The latter is a lot easier in the highly urban Northeast, West Coast, and Mid-Atlantic regions. However, if you work in sparsely populated areas like the Midwest, the Deep South, or much of Canada, you may not have the wireless coverage that you need. All is not lost in that case—because your audit software can be set up to work offline, and to transmit data back and forth when the users "sync" (connect) to their home PC, their office PC, or when they can get to a WiFi hub to deliver the data to the office (see "Explaining Wireless").

Do you want to print in the field? It is highly effective, and saves time and money, to print a report while the auditor is still at the customer site. But doing so requires a printer and appropriate software as well. Ask yourself whether you want to be able to do this, and what, for you, is a suitably sized report, before you get started purchasing hardware and software for printing.

Other extras? There are a wide variety of add-ons for PDAs today, from external keyboards to wireless earphone/headsets—for calls from the boss, not just for playing MP3s. Some other entry level extras are rubberized cases to protect the PDA from bumps and scratches; carrying cases that clip to a belt; and keyboards that roll up or collapse (see photo, p. 40).

But that's enough of the basics for this article. Let's move on to some of the new neat things that increase the power of your hand-held audit.

This advanced keyboard is actually a laser image projected onto a tabletop or any convenient flat surface.

Advanced Features

Getting different types of information into and out of your PDA more quickly is a focal point of new technology. Using this ability to help auditors capture data more quickly and accurately might be your starting point. Each of the following features either makes possible something that was not possible before (such as biometric security), or greatly enhances the current state of the art.

Template Drawings

The idea here is to provide a stock set of layouts for the kinds of homes or commercial structures that your auditors deal with most frequently. These layouts come in the form of a miniature blueprint graphic that is loaded into the PDA and is associated with the audit itself. The auditor can then sketch in notes about particular problem areas or other things that need to be remembered—things that either affect audit calculations or help your company to implement the measures that the auditor is recommending. (Note that sketching notes on the PDA interface requires a touch screen.)

Camera Input

A variation on template sketches is to provide camera (or even video) input and attach that to your audit. The auditor can visually record odd circumstances—which as they say, is worth 1,000 words. A further variation on this is to allow the auditor to capture the picture and then to add notes (a technique that John Madden made popular when he reported NFL football on late-night TV). The most recent PDAs come with a 3 megapixel camera, which is more than adequate for video capture, and pretty good for camera shots.

Explaining Wireless

There are three ways of being wireless. First, there are wireless cables that attach PDAs to peripheral devices, like printers. This is called a PAN (or Personal Area Network) connection. On a PDA, a PAN connection is usually accomplished with Bluetooth (see www.bluetooth.com). A PAN connection is usually made at less than 20 feet.

Then there is a WLAN (Wireless Local Area Network). This connection is made over WiFi and can connect at up to a few hundred feet. A WLAN is used to gain access to your local network, or even to the Internet and is used to pass data or files back and forth among these points. Stores that offer WiFi hubs are actually providing a WiFi access point (a hub) for you to connect to. WiFi hubs are also known as hotspots.

The widest ranging type of wireless connection is a WWAN (Wireless Wide Area Network). A WWAN covers a city or region and is offered by the wireless carriers, like Sprint. This type of connection allows you to connect to the Internet from anywhere within your coverage area; it is what is responsible for the lovely cell towers that you may see springing up in your town. Examples of WWAN implementations are EDGE and EV-DO.

Voice Input

The benefits of being able to record your voiced thoughts with the audit are similar to the benefits that doodling in thoughts provides. Like the camera, this gives the auditor a way to gather a great deal of data very quickly, especially when the data doesn't fit into the checkboxes and list boxes of the more formalized audit itself. The voice recording can be kept with the audit, on the PDA as well as back on the company server; or it can be converted to text to make it easy to search for terms and keep file sizes manageable.

Signature Fields

In a typical energy audit, the customer/homeowner is presented with a list of recommended measures (and the costs of implementing these measures). If appropriate, you may wish to have the customer sign off immediately on these recommendations, so that someone in your company can place an order for the required materials and the auditor can schedule the work with the customer. This cuts down a great deal on the work flow needed to implement the measures—meaning lower costs and faster response time for your customers.

A key piece is to get the customers' signature on the proposed work. The signature can be shown on a printed work order or it can be directly stored in the PDA, along with the audit and the work order. The latter approach eliminates the need for physical printers and all that goes with them, so the signing off can be completed in one trip. Furthermore, the digital acceptance signature is available to anyone in your systems—so that inventory can verify it, as well as accounts receivable; it's not just buried somewhere on a printed piece of paper.

Biometric Readers

Biometric readers provide fingerprint (or retinal) identification (see cover photo). This is a sure way to lock down the PDA or other devices to a particular user. Some PDAs come equipped with biometric readers already; others allow the user to install biometric readers if the PDA doesn't come equipped. Having a biometric reader reduces the risk of losing a PDA and the important data on it, if the data haven't been backed up somewhere else. If the PDA is not turned on with the biometric recognition, the data will remain encrypted, and may even be automatically deleted after a set period of time. The biometric reader checks for fingerprint patterning, much as police agencies track individuals by their fingerprints. When the would-be user presses the screen, light is emitted, and the resulting image is parsed to see if the person's fingerprint matches the fingerprint of one of the persons enabled for the device.

Radio Frequency Identification (RFID)

Think of RFID as a super, long-distance bar code reader. Inexpensive RFID tags send out electronic identification information and can be affixed to containers, palettes, materials, or machines. But while many bar codes can be printed with the same number, a RFID tag can be unique. This means that you can be sure of what the tag is attached to, assuming that the right tag is attached.

A RFID reader on your PDA allows you to pick up this information from a short distance (roughly 1 foot to 30 feet) away. This technology is especially valuable in commercial audit applications, where you may be inspecting and auditing machinery or vehicles.

Sensors

Today's PDAs can be relatively easily extended to accept input from external hardware or applications, such as blower door tests or temperature measurements. This information can be input into the PDA wirelessly, or through USB or serial cables.

Advanced Keyboards

There are advanced keyboards that are just light. One such "keyboard" is a laser image that is projected onto a tabletop (see photo, p. 42). As you move your fingers over the laser image of a key, that keystroke is entered. As this keyboard technology evolves, it will mean one less heavy item to carry around.

Voice-over Internet Protocol (VoIP)

Voice-over Internet Protocol is a way to make digital phone calls over the Internet. You do not need a voice plan for your phone to use VoIP, but you do need WiFi capability on your PDA device and access to a WiFi hub, and the person or company you are calling usually has to be a VoIP user. However, once everyone is set up this way, the technology works very inexpensively and pretty well nowadays. If long-distance calls or high phone bills are a concern of yours, VoIP deserves a look.

GPS

Global positioning receivers are showing up in a lot of the current generation of PDAs, and especially in phone-equipped PDAs. Having a GPS receiver on board allows you to use or create software that takes an address and tells the user exactly where he or she is in relation to that address. So if you are sending auditors to a series of homes, they can key in each address that they want, and the software will direct them to that address. This saves you from purchasing separate in-truck navigation systems, or the more likely desperate phone calls back to the central office asking for directions. (Coding the ability to do this directly into your audit application is no longer a difficult task either.)

The Only Constant Is Change

With this as a start, you are at least equipped with all of the right questions to ask before setting up your audit software. Don't be afraid of change in the industry; don't let fear prevent you from acting to automate your paper audits. Many of your competitors are going automated today. Change is a good thing in this industry and it is bringing you better and less expensive solutions than were available only one or two years ago.

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For more information:

To ask the author questions about energy software, visit his Q&A Web page, www.fo.com/QA, or contact him at bills@fo.com.

For more on the rules of thumb for user interface, go to www.foaudits.com/guidelines.aspx.

For audit demos and feature explanations, go to www.foAudits.com.

For the author's Handheld User Interface Ten Commandments, go to www.fo.com/articles/mc-01.htm.

For the latest mobile technology news, go to www.mobilitytoday.com.

To learn about Skype, a major VoIP player, go to www.skype.com.